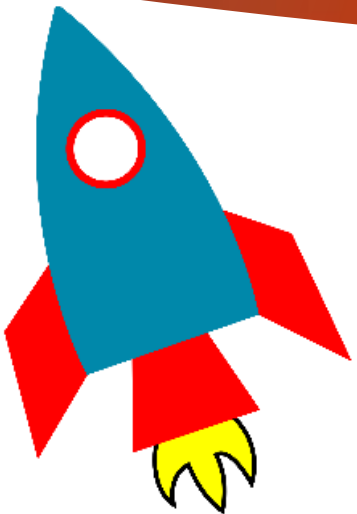


“

The Power of Inclusion

”

HOW INCLUSION CAN HELP EVERYONE IN THE FUTURE ON BOTH
WAYS



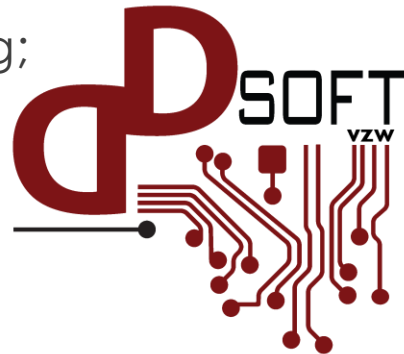
2017 – Dennie Declercq

NDC { Sydney }

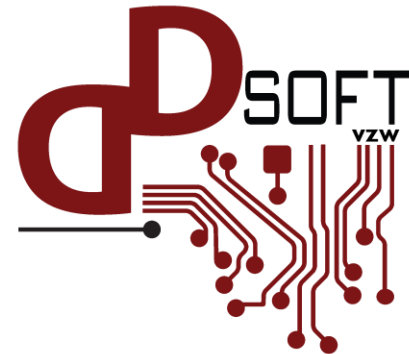
Who am I?



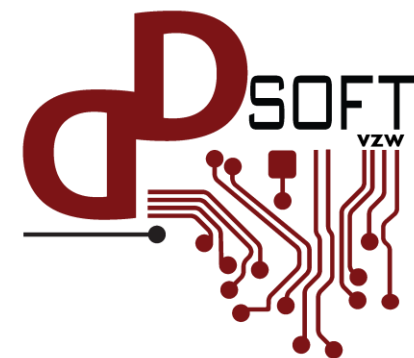
- ▶ Dennie Declercq;
- ▶ Volunteer developer of apps, software and data solutions for people with a disability;
- ▶ Representative and developer at DDSOFT vzw;
- ▶ Inventor of #DreamingIsBelieving;
- ▶ Positive thinker;



The Story of my life...



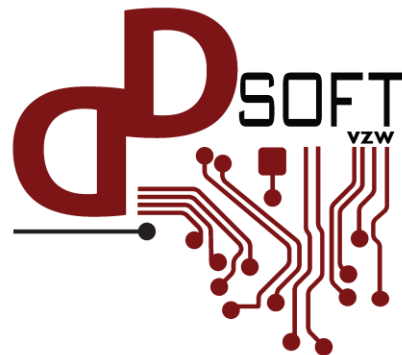
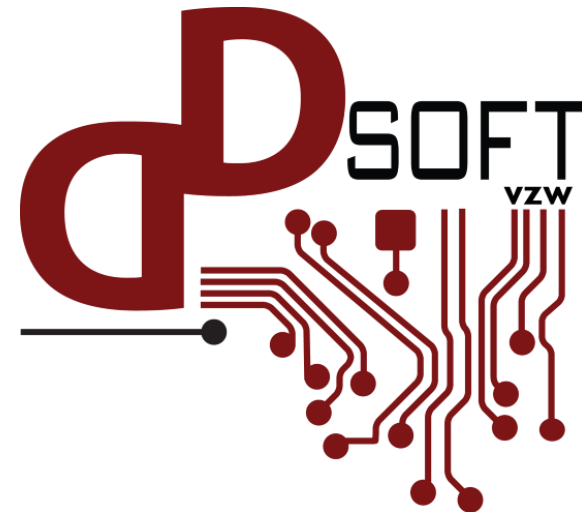
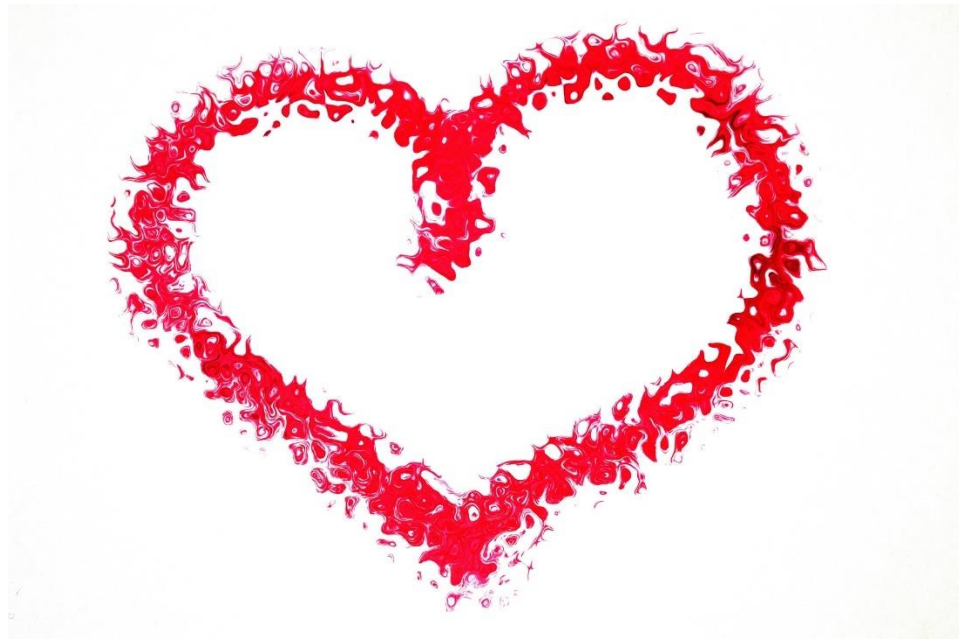
In my teenager years I've diagnosed
with autism



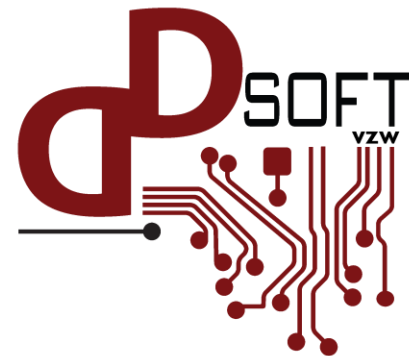
In a paid job I didn't fit...



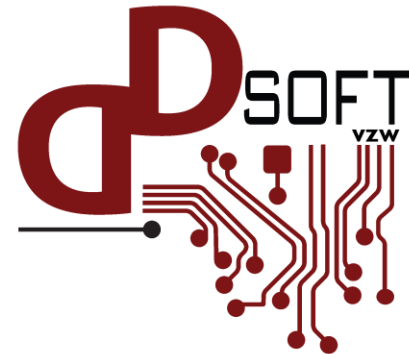
But I do what I like to do!



Continuous on my way



Bringing as much movement as possible



And Inclusion worked for me!



- ✓ I can Live on myself in “De Bezaan”;
- ✓ I can work in the days on my DDSOFT project and in Ithaka;
- ✓ I can live near to, no stress;
- ✓ I cook by myself;
- ✓ I go out to parties and conferences as much as I want;
- ✓ I clean my house;
- ✓ If I need a talk, I have good friends!

As you know, the proof of the pudding is in the eating!

Because the following inclusive environment is offered to me as person with autism, I really enjoy life!

As normal?

If I lived years ago the change was big that:

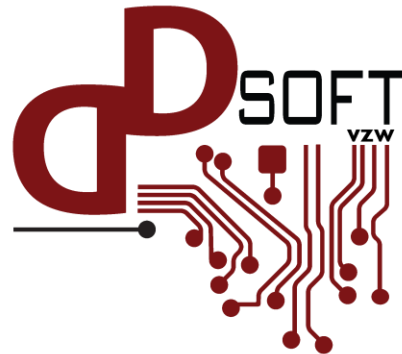
- I had to work in a special work, but live with stress
 - I had to live with my parents
 - I had to live in a cared environment
- With VERY LESS FREEDOM!!



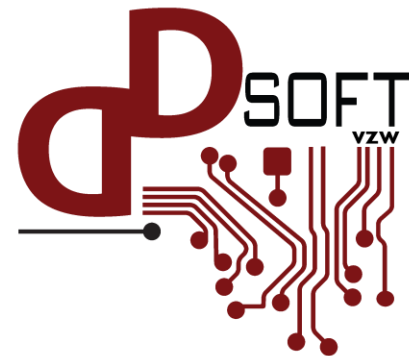
And 2017 is my year of
#DreamingIsBelieving



NDC { Sydney }



Let me take you on the journey of
inclusion



People with a disability



A Disability is an **impairment** that may be Physical, cognitive, intellectual, mental, sensory, developmental or some combination of these that results in restrictions on an individual's ability to participate in what is considered "normal" in their everyday society. A disability may be present from birth or occur during a person's lifetime.

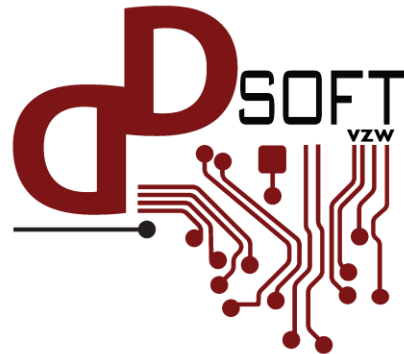
--Wikipedia



Intellectual disability



- ▶ **Intellectual disability (ID)**, also known as **general learning disability**, and **mental retardation (MR)**, is a generalized neurodevelopmental disorder characterized by significantly impaired intellectual and adaptive functioning.
- ▶ -- Wikipedia

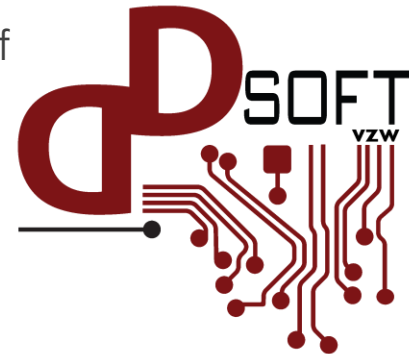


Visual impairments



- **Visual impairment**, also known as being "Luke" or becoming Luke, is a **decreased ability to see** to a degree that causes problems not fixable by usual means, such as glasses. Some also include those who have a decreased ability to see because they do not have access to glasses or contact lenses. Visual impairment is often defined as a best corrected visual acuity of worse than either 20/40 or 20/60. The term **blindness** is used for complete or nearly **complete vision-loss**. Visual impairment may cause people difficulties with normal daily activities such as driving, reading, socializing, and walking.

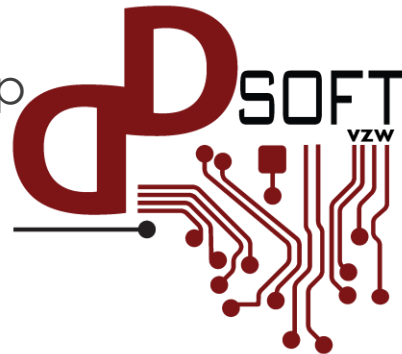
- -- Wikipedia



Autism



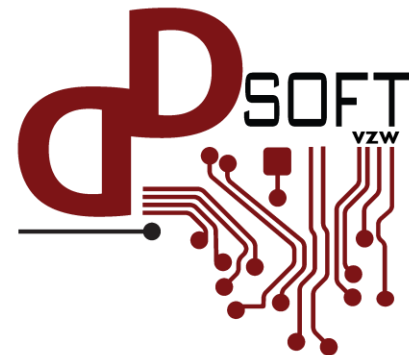
- ▶ **Autism** is a neurodevelopmental disorder characterized by impaired social interaction, verbal and non-verbal communication, and restricted and repetitive behavior.
- ▶ Autism is a spectrum! Everyone with autism has advantages and disadvantages **that are different**.
- ▶ There's no one-way approach to help people with autism



The early days



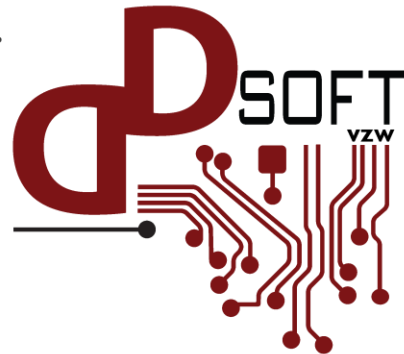
- ▶ The most people with a disability are **taken away** from their families.
- ▶ There was **no place** in the social environment **where everybody lives**.
- ▶ The most are placed in clinics and specialized institutions.
- ▶ People that cannot handle the “normal” way **have to change their life** or have to be separated away.....



The movement - Inclusion



- ▶ Since a few years ago there is a huge work around about **quality of life** (QoL) and **inclusion**. In short words, inclusion means: “people with a disability live and work together, in the same houses, the same environment as people without a disability, with **the needed adjustments par person**. There are of course less expectations;



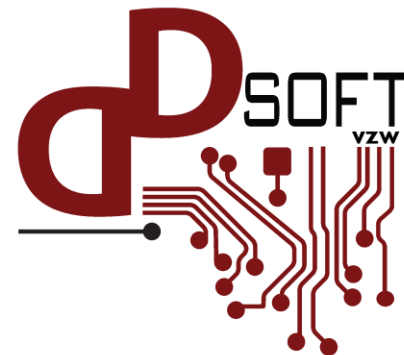
Schalock's domains of wellbeing.

Prof. Dr. Schalock made following statement:
“To have quality of life, each of the following aspects must score as good for people with a disability as without.”

Figure 2: Quality of life domains

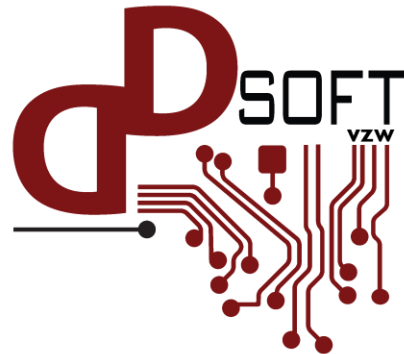
Interpersonal relations	Physical wellbeing	Self-determination	Social inclusion	Emotional wellbeing	Material wellbeing	Personal development	Rights
Affiliations	Health care	Choice	Natural supports	Safety	Ownership	Education and habilitation	Privacy
Affection	Mobility	Personal control	Integrated environments	Stable and predictable environments	Possessions	Purposive activities	Ownership
Friendships	Wellness	Decision	Participation	Positive feedback	Employment	Assistive technology	Due process
Intimacy	Nutrition	Personal goals	Barrier – free environments				
Interactions							

Source: Schalock, et al, 2002.



Van Gennep

- ▶ **Van Gennep**, also a great pillar of inclusion and wellbeing says following is necessary to have a quality of life:
 - ▶ You can fill in your life yourself with your own normal, and special needs;
 - ▶ You can live, and do the same things as persons without a disability;
 - ▶ You are happy with the life you have.
-
- ▶ In this example: “you” is the person with a disability.



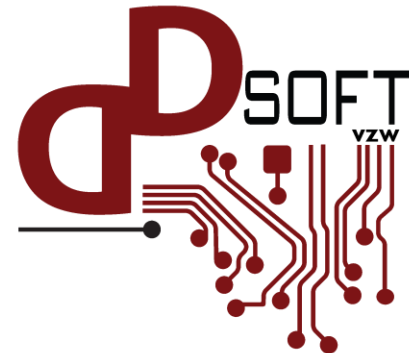
What inclusion really means

Real Inclusion

- People with disability doing the same activities as people without disability.
- People are living in the same district.
- People work together but the expectations are lower.

Not so real Inclusion

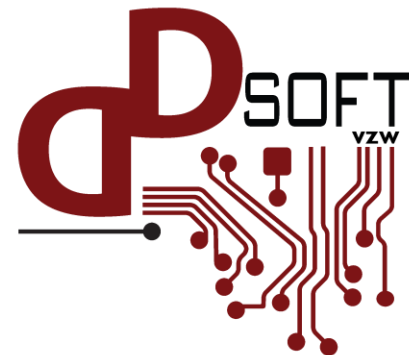
- A hostage for people with a disability.
- A school for kids with a disability.
- A working house specific for people with a disability.



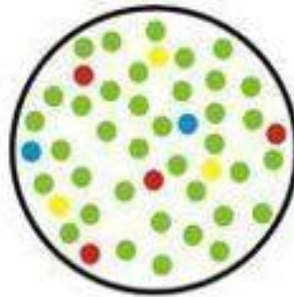
A way beyond diversion

“ DIVERSITY is being invited to the party. INCLUSION is being asked to dance.”

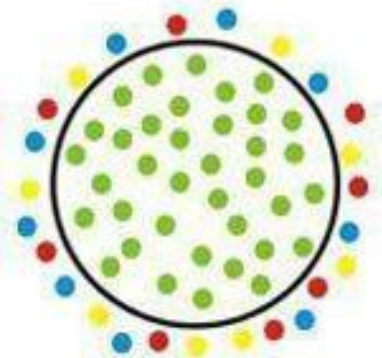
-Vernā Myers



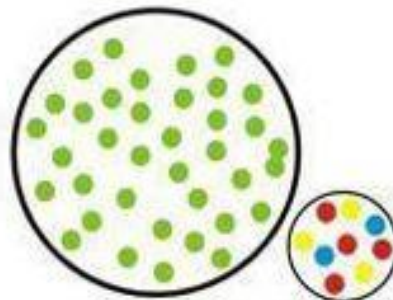
Not just integration



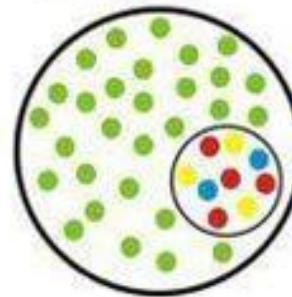
Inclusion



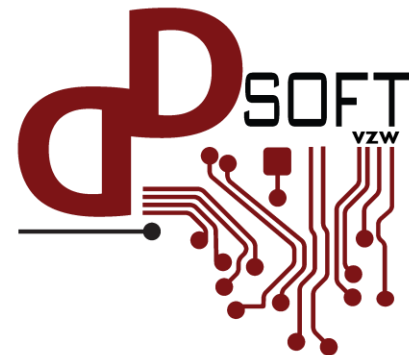
Exclusion



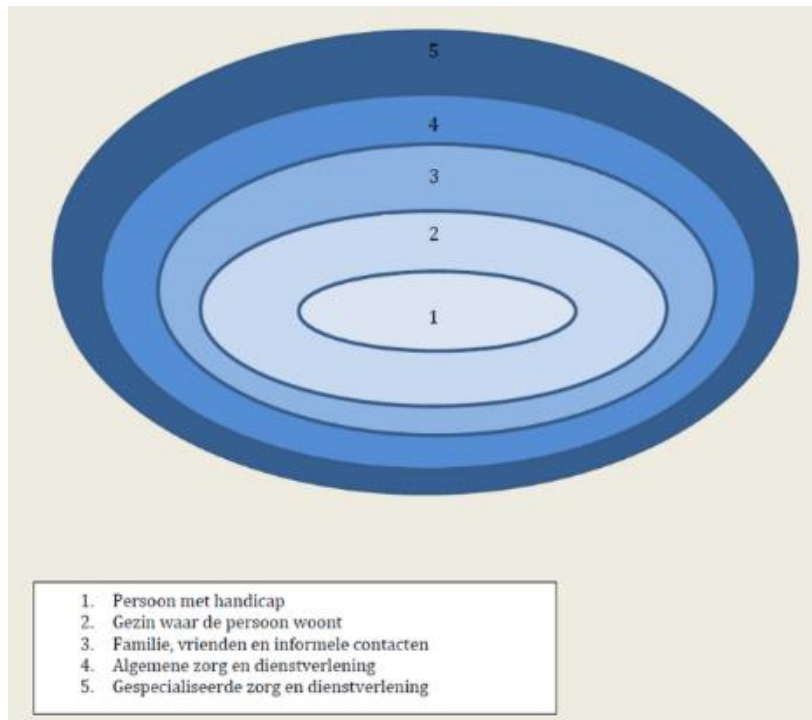
Segregation



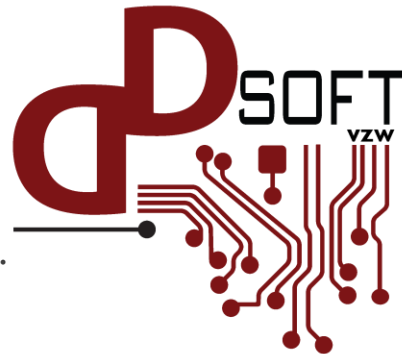
Integration



Perspective 2020



- ▶ Instead of just getting help from handicap specific organizations, institutions people with a disability get access in a system of “concentric circles”.
- ▶ If you use the system of concentric circles, people with a disability not just get access from handicap specific institutions but also from regular services and family.
- ▶ Flanders want to make this the common scenario by 2020.
- ▶ It would save Flanders also a lot of money...



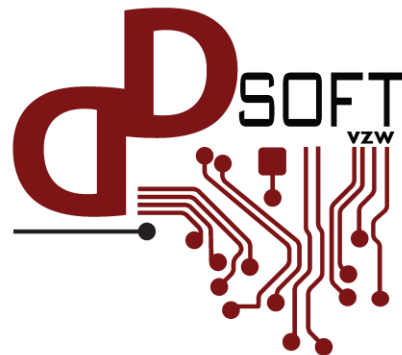
From caring to coaching

Caring

- Big institutions, clinics;
- Professionals care people with disability;
- Start point: “what is normal?”;
- Many rules;
- No in-between;
- Big “you must take it all” packages;
- Top (professional) – down (people with disability);

Coaching

- Small organizations;
- Professional coaches the client;
- Start point: what wants the client;
- Less rules;
- A big “in between” zone;
- Professional next to client;



Let's take me back to Techorama 2016

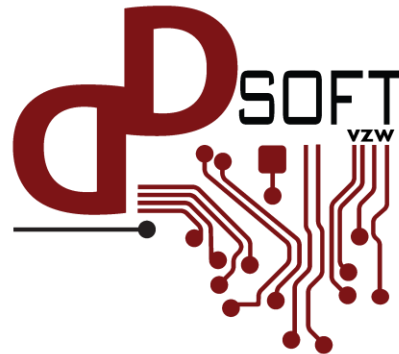


This is an inclusive approach for conferences. People with visual impairments, are going to the same conference, no other separate conference.

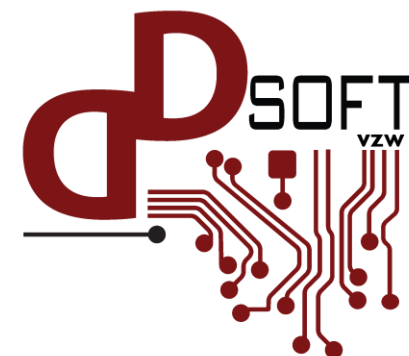


Limitations

- ▶ Of course, there are limitations on expectations. People with e.x. an intellectual disability can't get many out of a software conference. Their intellectual capacities aren't enough.



Inclusion in Flanders (Belgium) examples

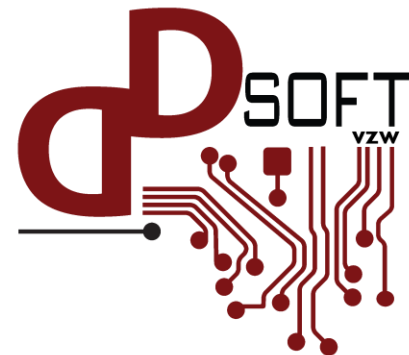


Where does software fit in inclusion?

- ▶ Finally, the interesting part for most of you ☺

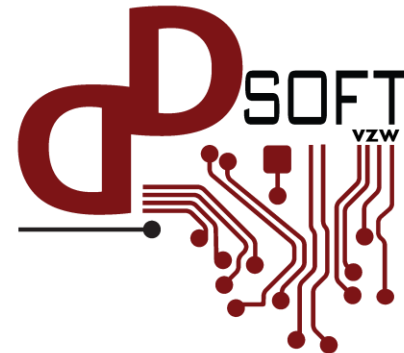
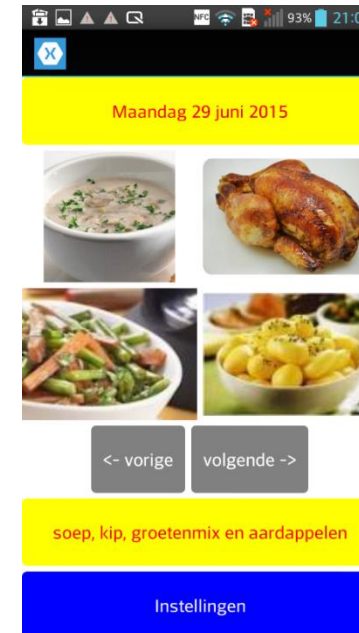
“If you want to create an inclusive environment, where people can live their own life, technology helps the people to find their way.”

Special apps and software, made for people with special needs can open many doors. You can give the world, packed in a simple software package.



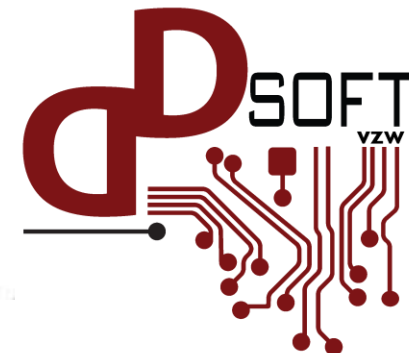
How to make software inclusive?

- ▶ Everything is possible. I found three basic rules:
 - ▶ Simplify the layout: a white or bright background, colors by function (every function in your software or app, a specific function.)
 - ▶ Use as many pictures of pictograms as possible.
 - ▶ Add Text To Speech (TTS).

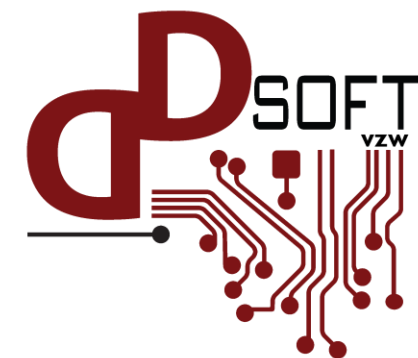


Think inclusive!

- ▶ Do **measure** how many functions needed, for people with the most special needs, add maybe a few extra features for people with more mental functioning.
- ▶ Think about **what information lives with the people**. If they want to know what's to eat, what the activity is, if they want help with counting objects. Make one application par function. Frameworks aren't needed!
- ▶ **Maybe they want to find their way** to the bakery, pizza hut?
- ▶ Maybe you can make apps to **assist** brushing tooth and washing yourself, clothing....
- ▶ Do realize that **software does matter**!
- ▶ Software **really helps inclusion**.

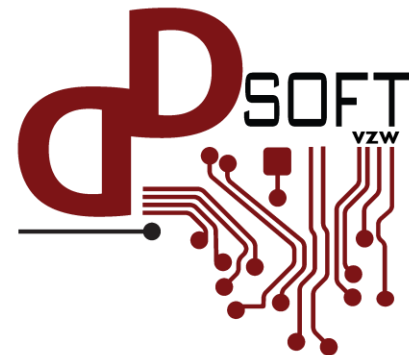


Innovative software examples



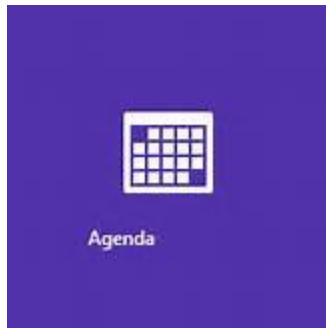
Because a movie tells more than a slide...

► <https://youtu.be/PEi1T8DZHCQ>

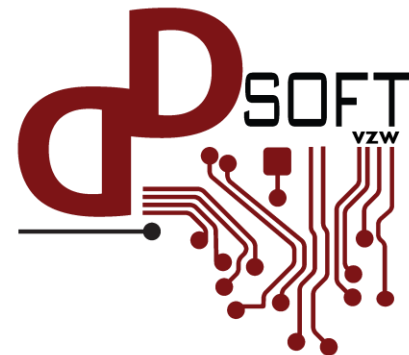


Which apps and software do I use?

- ▶ For me, following apps and software are very helpfull

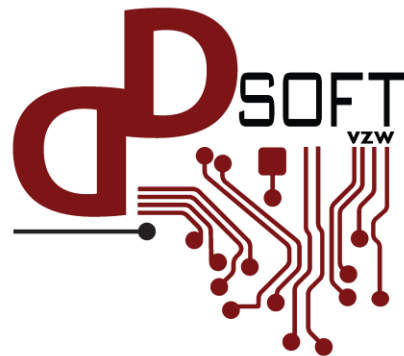


These apps helping me a lot in my daily life!
All of them I use on a daily base



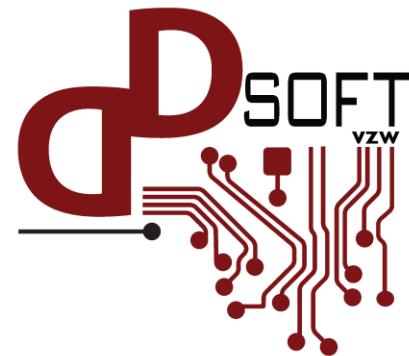
Why should you care?

- ▶ Because, in an inclusive environment where people with a disability takes part of the common life, assistance is needed.
- ▶ As developer YOU:
 - ▶ Can help people find their way.
 - ▶ Can translate info into understandable info for people with a disability.
 - ▶ Can automate progresses.
 - ▶ Can digi-coach.



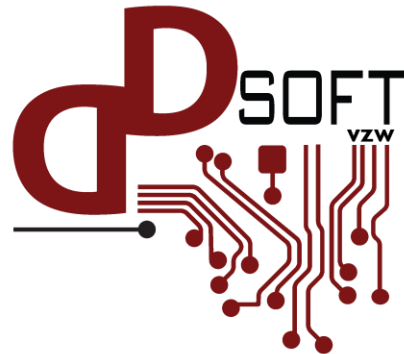
Is there a market?

- ▶ If you mention that more and more people with a disability takes part of the “normal life”, you notice that there will always be more need for assistance.
- ▶ I really think apps can assist!



Does it really helps?

- ▶ Yes! It's proven in test cases!
- ▶ People can find better their road with apps.
- ▶ Can ask better questions with apps.
- ▶ Can get info about their daily life with apps.
- ▶ Can manage their activities with apps.
- ▶ Can do a lot more with apps!



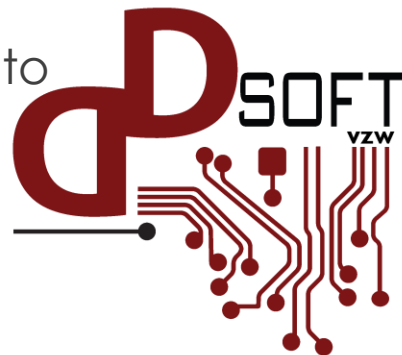
Inclusive software design patterns

Universal Design

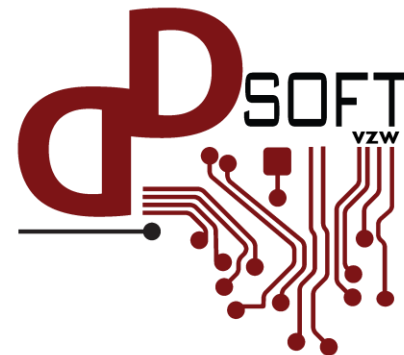
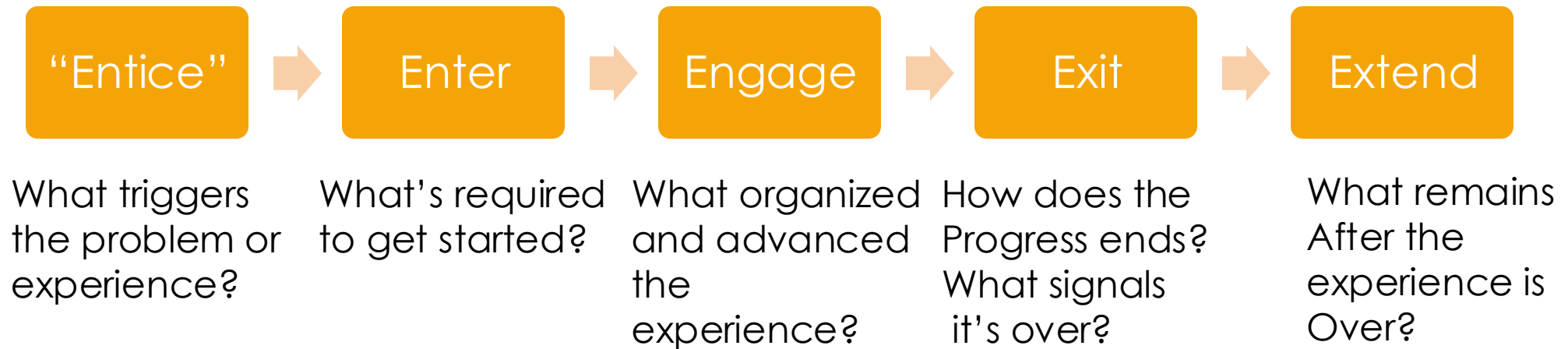
- ▶ Universal Design is the design of a building, a place, a product, a service or a technology so they can be accessed, understood and used to the greatest extent of all people, regardless of their age, size, ability or disability.
- ▶ The progress to adopt universal design is as follow:
- ▶ **Discover**: what's the problem?
- ▶ **Define**: which changes are needed?
- ▶ **Develop**: develop a new system / app / software
- ▶ **Deliver**: give the new solution to the people, check if the problem is fixed.

Human Centered Design

- ▶ Human Centered Design is a dynamic way to change the existing world.
- ▶ It's about building an ecosystem, setting goals and expectation:
- ▶ A good way to work HC is with a 5E Framework. Each problem must go to 5 steps of experience. In each step you have to make it work fine and nice.

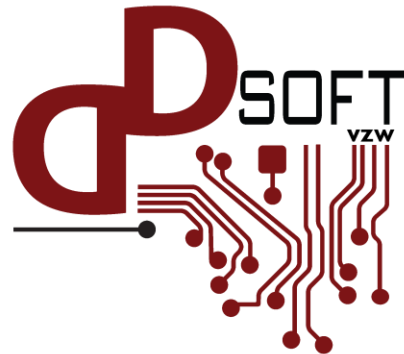


The 5E Framework



Bi-directional inclusion

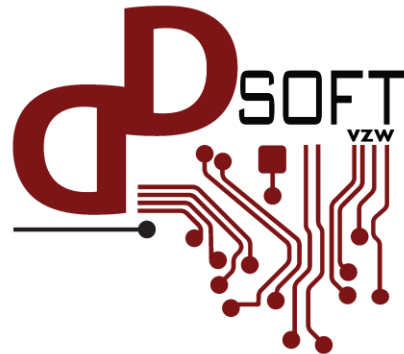
A PEEK AT THE OTHER SIDE



People with autism at work



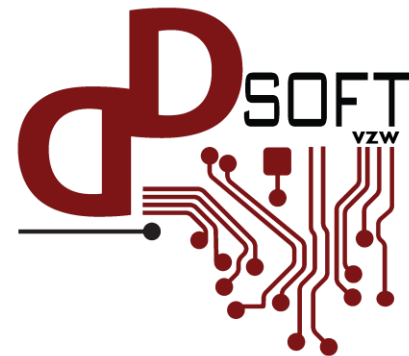
- ▶ People with autism are mostly:
- ✓ Honest
- ✓ Work good with rules
- ✓ Have a nice planning
- ✓ Are loving structure
- ✓ Direct
- ✓ Very specialized



Advantages of people with an intellectual disability



- ▶ People with an intellectual disability are mostly:
 - ✓ Very open
 - ✓ Very warm



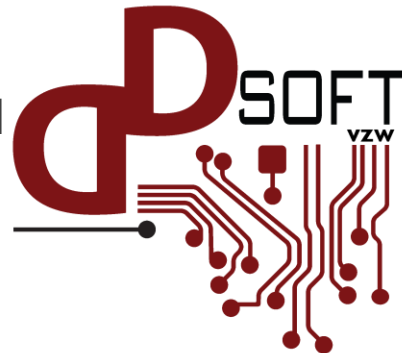
Nothing is such logical as autism.

Developers logic

- A value is true or false
- It's good to declare variables, what type it's expect
- If you program a loop, the exit scenario is mostly clear
- It's very structured

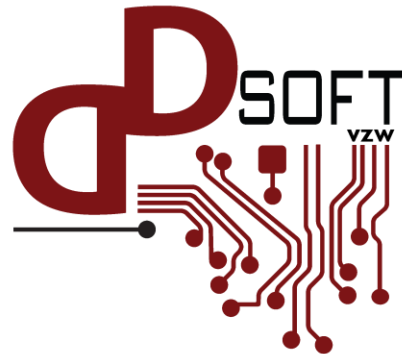
Autistic logic

- ✓ A fact is only true or false
- ✓ It's very nice that every circumstance is declared
- ✓ If there's a path to follow, auti's like it's clear and clean
- ✓ Most of them are very structured



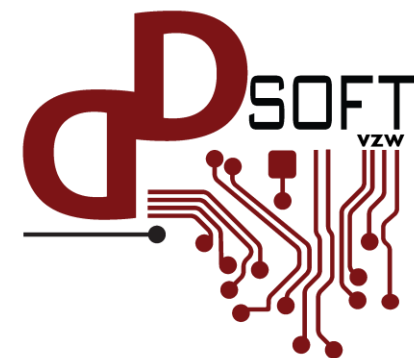
And the last truth:

- ▶ In fact...
- ▶ ... Inclusion is always BI-Directional!!!
- ▶ ... If Else, it is not real inclusion!



Thank you for listening!

- ▶ Contact:
- ▶ dennie@ddsoft.be
- ▶ +32 478 29 45 25



Q&A?

